Towards Business Intent Ontological Unpacking in Decision-Making

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Abstract: Decision-making in organizational environments frequently begins with business intent expressed through heterogeneous textual artifacts, such as policies, expert guidelines, regulatory documents, and analytical questions. Although rich in domain knowledge, these artifacts are typically informal and semantically underspecified, creating a gap between business intent and the explicit structures required for grounded computational reasoning. This paper introduces an ontology-guided methodology for *Business Intent Ontological Unpacking*, in which AI-mediated semantic analysis reconstructs the implicit ontological commitments underlying such artifacts. The extracted structures are formalized into a *Business Intent Model (BIM)*, an executable semantic artifact that captures intent semantics, structural commitments, evidential grounding, and reasoning structures. The methodology is evaluated in a cybersecurity decision-support scenario involving organizational contract criticality assessment, including qualitative expert assessment, comparison with a retrieval-based enterprise question-answering baseline, and executable reasoning validation. Results indicate that the proposed approach consistently preserves decision-relevant semantic structures while enabling deterministic, traceable, and explainable reasoning. These findings suggest that ontology-guided semantic compilation offers a promising foundation for grounded enterprise decision-support systems.

1 Introduction

Decision-making in organizational environments frequently begins with business information needs expressed in natural language. Domain experts, analysts, and decision-makers articulate objectives, concerns, policies, and evaluative criteria through textual descriptions, questions, procedural guidelines, or regulatory statements. These artifacts externalize an underlying Business Intent that guides analytical investigation and decision-making. However, because such intent is typically expressed in concise and semantically underspecified forms, critical analytical assumptions often remain implicit, making computational interpretation difficult.

This challenge becomes particularly significant in scenarios where effective decision support depends on aligning business intent with heterogeneous organizational evidence. Consider cybersecurity risk assessment in organizational contracts: decision criteria may be described informally in policy documents or expert guidelines, while the factual evidence required for evaluation resides in structured repositories such as contract databases, operational systems, or organizational records. Connecting these heterogeneous representations requires making explicit the semantic commitments embedded in the original business intent, including relevant domain entities, evaluative distinctions, semantic dependencies, and the real-world conditions that justify decision outcomes.

Ontology-driven conceptual modeling offers a principled foundation for addressing this problem. In this paradigm, conceptual models are understood not merely as data structures, but as explicit representations of ontological commitments concerning the entities, relations, events, and distinctions that characterize a domain. Recent work has introduced the notion of *ontological unpacking*, in which symbolic representations are explained by revealing the ontological commitments and truthmakers that make their statements meaningful. This perspective transforms descriptive representations into explanatory semantic artifacts capable of supporting semantic clarity and grounded reasoning.

Despite these advances, existing research has focused primarily on unpacking conceptual models after they have already been constructed. Much less attention has been devoted to the earlier and equally important stage in which business intent itself must be reconstructed from informal textual artifacts. In practice, decision-support systems often begin not with formal conceptual models, but with heterogeneous descriptions of requirements, policies, questions, and domain criteria expressed by stakeholders. Transforming these artifacts into explicit semantic representations remains largely a manual and ad hoc pro-

cess.

At the same time, recent advances in artificial intelligence, particularly large language models (LLMs), create new opportunities for mediating this transformation. LLMs have demonstrated strong capabilities for extracting semantic structures from unstructured text, but most applications emphasize information extraction, retrieval, or question generation rather than the systematic reconstruction of ontological commitments underlying business intent.

To address this gap, this paper proposes an ontology-guided methodology for *Business Intent Unpacking*, operationalized as an AI-mediated process for transforming heterogeneous textual artifacts into an explicit *Business Intent Model (BIM)*. The BIM is conceived as an executable semantic representation of business intent, capturing domain concepts, semantic relations, evaluative criteria, operational question structures, and explicit traceability to supporting evidence. Rather than treating natural-language requests as direct inputs for retrieval or answer generation, the proposed approach reconstructs their implicit semantic commitments as a formal substrate for grounded reasoning.

This work investigates the following research questions:

- **RQ1:** To what extent can ontology-guided AI-mediated unpacking transform semantically underspecified business intent into semantically adequate Business Intent Models?
- **RQ2:** How does the proposed BIM-based approach compare with retrieval-based question-answering architectures in terms of semantic grounding, traceability, and decision-support adequacy?
- **RQ3:** Can the generated Business Intent Models support executable decision queries over organizational evidence in an operationally feasible and reusable manner?

The contributions of this paper are threefold. First, we introduce Business Intent Unpacking as a methodology that operationalizes ontological unpacking at the level of decision-support requirements rather than post hoc conceptual analysis. Second, we define the Business Intent Model as an explicit executable semantic artifact capable of formalizing implicit business intent with evidential traceability. Third, we provide an empirical evaluation combining expert qualitative assessment, comparison with a retrieval-based baseline, and executable decision-support validation in a cybersecurity contract assessment scenario.

The remainder of this paper is organized as follows. Section 2 presents the theoretical foundations of ontological commitments, foundational ontologies, and ontological unpacking. Section 3 reviews related work in ontology-driven conceptual modeling, requirements formalization, and retrieval-based decision support. Section 4 presents the proposed methodology and formally defines the Business Intent Model. Section 5 describes the implementation architecture and operational realization of the approach. Section 6 presents the empirical evaluation and comparative analysis. Finally, Section 7 concludes the paper and outlines future research directions.

2 Background

Conceptual models play a central role in the design and operation of information systems, as they provide structured representations of domain knowledge used for data integration, reasoning, interoperability, and system development (Guarino et al., 2020; Proper and Guizzardi, 2021). More broadly, this perspective aligns with classical ontology engineering, where ontologies are understood as explicit specifications of shared conceptualizations intended to support semantic interoperability across computational systems (Gruber, 1995; Guarino, 1998). These models typically appear in the form of schemas, knowledge graphs, ontologies, or logical specifications describing entities, relations, constraints, and events within a domain.

A central insight in ontology-driven conceptual modeling is that symbolic representations such as schemas, knowledge graphs, or logical models inevitably make assumptions about the nature of the domain they represent. These assumptions correspond to what philosophy refers to as the *ontological commitment* of a representation, that is, the kinds of entities and relations that must exist in the world for the statements expressed in that representation to be true (Quine, 1948). According to (Guarino et al., 2020), understanding the meaning of conceptual models therefore requires going beyond purely formal or logical semantics toward *real-world semantics*, which concerns the mapping between symbolic structures and the entities, properties, and relations that exist in the modeled domain (Guizzardi, 2020; Wieringa, 2011). This perspective is aligned with broader knowledge representation traditions, where semantic structures are understood not merely as symbolic encodings, but as formal representations of domain meaning intended to support reasoning and explanation (Brachman and Levesque, 2004).

The process of making these ontological commitments explicit is known as *ontological analysis*. On-

tological analysis seeks to identify the categories of entities that exist in a domain, their properties, and the relations that connect them. This analysis is typically guided by foundational ontologies, such as the Unified Foundational Ontology (UFO), which provide domain-independent distinctions concerning objects, events, roles, relations, dispositions, and relators (Guizzardi et al., 2022). Within this context, (Guizzardi and Guarino, 2023) introduces the notion of *ontological unpacking* as a specific explanatory process applied to symbolic models. Ontological unpacking reveals the ontological commitments underlying a conceptual model by identifying the entities in the domain that act as the *truthmakers* of the propositions expressed in that model. By making explicit the structures underlying symbolic representations, ontological unpacking clarifies the meaning of concepts and relations, reduces ambiguity, and transforms conceptual models from descriptive artifacts into explanatory semantic structures (Guizzardi and Guarino, 2023).

This perspective is particularly relevant in contexts where semantic clarity and interoperability are critical. In modern data ecosystems, information is often distributed across heterogeneous systems that employ different schemas, terminologies, and modeling assumptions. Achieving semantic interoperability in such environments requires understanding how the conceptualizations embedded in different systems relate to one another (Guizzardi, 2020). More generally, conceptual modeling research emphasizes that interoperability depends not only on structural compatibility, but also on semantic alignment between domain constructs across organizational contexts (Krogstie, 2012). Ontological unpacking therefore provides a methodological basis for identifying hidden assumptions, resolving ambiguities, and aligning domain models. It also connects naturally with explanatory traditions in philosophy and artificial intelligence, particularly through notions of grounding and truthmaking (Moltmann, 2007). In ontology-driven conceptual modeling, this perspective is operationalized through modeling patterns derived from foundational ontologies. For example, (Fonseca et al., 2019) show that many relations appearing in conceptual models are in fact derived from more fundamental ontological structures, such as relators, events, or dispositions.

Importantly, ontological unpacking is not tied to any specific application domain. Instead, it provides a general methodological basis for transforming informal or underspecified conceptual representations into explicit ontological models, supporting semantic clarification, interoperability, knowledge representation, and explainable reasoning. In many practical scenarios, conceptual models are constructed to support the exploration of domain knowledge in response to specific information needs, often expressed as competency questions guiding ontology development (Grüninger and Fox, 1995). These needs are frequently formulated as questions posed by users or decision-makers, which implicitly rely on assumptions about the underlying domain (Van Fraassen, 1977). More broadly, requirements engineering has long recognized that stakeholder intentions, goals, and constraints must be explicitly modeled before computational systems can reliably support decision-making (Van Lamsweerde, 2001).

In this work, we adopt ontological unpacking as the conceptual foundation for structuring business intent into explicit executable semantic representations. Building on these theoretical foundations, the methodology presented in this paper operationalizes ontological unpacking as a systematic AI-mediated process for transforming implicit business requirements into explicit Business Intent Models capable of supporting grounded reasoning and decision-support processes.

3 Related Work

Related work in this area spans complementary research streams, including ontology-driven conceptual modeling and ontological unpacking, ontology engineering and competency-question–driven requirements elicitation, ontology learning from textual artifacts, and explainable symbolic reasoning for intelligent decision support.

In ontology-driven conceptual modeling, substantial effort has been devoted to revealing the hidden semantics of conceptual models through ontological analysis grounded in foundational ontologies (Fonseca et al., 2019). This line of work has clarified the ontological nature of modeling constructs such as relations, events, roles, and categories, particularly in the context of the ER conference series and related venues (Almeida et al., 2019). Representative examples include the ontological analysis of relations in conceptual modeling, the treatment of events as first-class ontological entities, and the use of foundational distinctions to improve semantic interoperability and conceptual clarity (Guarino and Guizzardi, 2015). More recently, this line evolved into explicit work on ontological unpacking, where symbolic domain descriptions are explained by exposing their ontological commitments and underlying truthmakers. (Guizzardi et al., 2021) on the viral conceptual model is a landmark in this direction, showing how unpack-

ing transforms descriptive models into explanatory ones, while subsequent work generalized the relation between explanation, semantics, and ontology (Guizzardi and Guarino, 2024).

A closely related line of research concerns the explicit treatment of information needs during ontology development. Competency Questions have long been used as a way to capture the functional requirements of ontologies, delimit scope, identify relevant concepts, and support validation. Early work such as Ren et al. showed how competency-question patterns can drive ontology authoring and semantic requirements specification, while more recent studies examined their broader role in ontology engineering practice (Ren et al., 2014; Monfardini et al., 2023; Keet and Monfardini, 2024). This literature is especially relevant because it treats natural-language questions not merely as informal prompts, but as structured expressions of what an ontology should represent and support. Closely related, goal-oriented requirements engineering has long emphasized the need to explicitly model stakeholder intentions, goals, and constraints before system realization (Van Lamsweerde, 2001). However, these approaches generally focus on requirement specification rather than the semantic reconstruction of business intent into executable reasoning structures.

Another relevant research direction concerns ontology learning and semantic structuring from textual artifacts. Classical ontology learning approaches explored the automatic derivation of ontological structures from unstructured corpora through linguistic analysis and statistical extraction (Maedche and Staab, 2001; Cimiano and Völker, 2005). More recently, AI-assisted ontology engineering has revisited this direction through large language models and knowledge graphs, including efforts to automate the generation of competency questions from normative and technical documents (Di Nunzio et al., 2024). These approaches demonstrate the feasibility of transforming textual artifacts into structured semantic representations. However, they primarily focus on concept extraction, ontology acquisition, or question generation, rather than on systematically reconstructing the deeper ontological commitments underlying business intent.

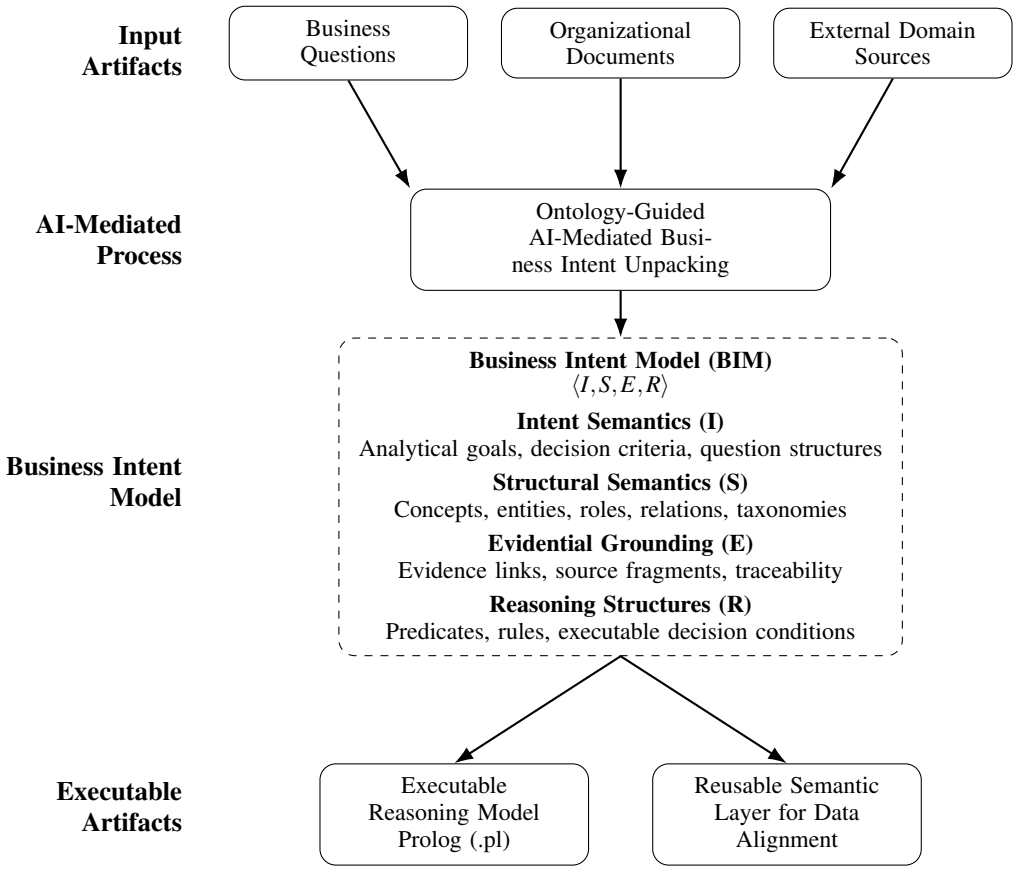
Explainable symbolic reasoning and neuro-symbolic AI also provide an important adjacent perspective. Recent work has emphasized the importance of combining learned representations with explicit symbolic reasoning structures to improve interpretability, controllability, and explainability in intelligent systems (Garcez et al., 2019). Similarly, formal decision modeling approaches such as Semantic DMN demonstrate how explicit logical representations can support explainable decision automation in the presence of structured domain knowledge (Calvanese et al., 2019). Ontology-Based Requirements Engineering offers an additional bridge between conceptual clarification and practical requirements analysis, where ontological analysis is directly applied to the elicitation and analysis of requirements. Recent work on ethicality requirements shows that ontological unpacking can guide requirements elicitation, ontology instantiation, and downstream analysis, effectively connecting conceptual clarification with operational requirements models (Souza et al., 2023). Together, these approaches reinforce the importance of explicit executable semantic artifacts, but generally assume that the relevant decision structures have already been formally specified.

Despite these advances, an important gap remains in the state of the art. Existing work on ontological unpacking has focused primarily on explaining conceptual models and improving semantic interoperability; competency-question and requirements-engineering research has focused on capturing information needs and validating ontologies; ontology learning has focused on extracting semantic structures from text; and neuro-symbolic reasoning approaches typically assume pre-existing formal knowledge structures. What remains largely unexplored is an integrated methodology that starts from heterogeneous natural-language business artifacts, performs AI-mediated ontological unpacking over these inputs, and produces explicit executable semantic representations that preserve ontological commitments, truthmakers, decision criteria, semantic dependencies, and evidential traceability. The methodology proposed in this paper addresses precisely this gap by connecting ontological unpacking, AI-mediated semantic reconstruction, and executable decision-support reasoning through the formalization of Business Intent Models.

4 Methodology

The methodology proposed in this work operationalizes ontological unpacking as a systematic AI-mediated process for transforming semantically underspecified business intent into explicit executable semantic representations. The central premise is that decision-support requests articulated by domain experts, analysts, or organizational stakeholders rarely express all semantic assumptions required for grounded reasoning. Instead, they externalize only a partial representation of the underlying analytical intent, leaving critical domain distinctions, evaluative

Figure 1: Ontology-guided methodological process for transforming business-oriented textual artifacts into an executable Business Intent Model.



criteria, semantic dependencies, and evidential conditions implicit. The proposed methodology addresses this problem by reconstructing these latent ontological commitments and organizing them into an explicit Business Intent Model (BIM) capable of supporting executable reasoning and explainable decision-support processes.

Figure 1 summarizes the proposed methodology. The process begins with heterogeneous business-oriented textual artifacts, including analytical questions, regulatory texts, institutional policies, technical reports, procedural guidelines, and domain documentation. These artifacts collectively express the business intent that motivates downstream analytical reasoning.

4.1 Ontology-Guided Business Intent Unpacking

The first methodological stage consists of ontology-guided semantic unpacking mediated by Large Language Models (LLMs). In accordance with the theoretical foundations presented in the Background section, ontological unpacking seeks to reveal the real-world entities, relations, and conditions that act as truthmakers for the propositions embedded in the original business-oriented artifacts. Rather than treating textual descriptions as isolated prompts for retrieval or direct answer generation, the methodology interprets them as partial manifestations of deeper semantic commitments that must be reconstructed before computational reasoning can be reliably performed.

During this stage, the LLM performs a structured semantic analysis of the input artifacts guided by ontological distinctions derived from foundational conceptual modeling principles. In this work, we adopt the Unified Foundational Ontology (UFO) as the ontological reference framework due to its well-established treatment of domain-independent distinctions such as objects, events, roles, relations, dispositions, and relators. However, the proposed methodology is not intrinsically tied to UFO and can, in principle, be instantiated with alternative foundational or domain ontologies depending on the decision-support

context. Under this guidance, the semantic analysis identifies domain concepts, entities, semantic relations, evaluative categories, decision criteria, taxonomic distinctions, competency-question structures, and evidential dependencies that collectively characterize the semantic structure of the underlying business intent.

Unlike traditional ontology learning approaches focused primarily on concept extraction, the proposed methodology seeks to reconstruct executable semantic commitments suitable for downstream reasoning. As a result, the output is not merely a descriptive ontology fragment, but a structured semantic artifact explicitly aligned with decision-support requirements.

4.2 Business Intent Model

The central formal artifact produced by the methodology is the *Business Intent Model (BIM)*. Conceptually, the BIM acts as an intermediate semantic layer between informal business-oriented textual artifacts and downstream analytical reasoning over organizational evidence.

Formally, we define the BIM as:

$$BIM = \langle I, S, E, R \rangle$$

where:

- I represents the *Intent Semantics Layer*, capturing the explicit analytical goals, evaluative distinctions, business criteria, and competency-question structures that define the intended decision-support objective;
- S represents the *Structural Semantic Layer*, capturing the ontological commitments reconstructed during unpacking, including domain concepts, entities, roles, taxonomies, and semantic relations;
- E represents the *Evidential Grounding Layer*, preserving explicit traceability between reconstructed semantic commitments and the textual evidence from which they were derived;
- R represents the *Reasoning Layer*, containing executable logical structures, predicates, inferential rules, and operational decision conditions that support computational reasoning.

This formulation emphasizes that the BIM is not merely a descriptive semantic artifact, but an executable semantic commitment structure explicitly designed to support explainable reasoning over organizational evidence.

4.3 AI-Mediated Semantic Compilation

The construction of the BIM is operationalized through a constrained AI-mediated semantic compilation process. Large Language Models are employed not as autonomous answer generators, but as semantic mediators responsible for transforming heterogeneous textual artifacts into normalized semantic structures under explicit ontological guidance.

To improve consistency and reduce semantic drift, the extraction process follows canonical semantic categories aligned with the BIM structure. The LLM is required to explicitly identify extracted concepts, relations, criteria, and evidential justifications, ensuring that generated semantic commitments remain traceable to the source artifacts. This constraint-based mediation reduces the likelihood of purely speculative generations and increases semantic consistency across repeated unpacking cycles.

This design choice is particularly important because business intent frequently spans heterogeneous and partially redundant documentation, requiring semantic synthesis rather than simple extraction from isolated textual fragments.

4.4 Executable Reasoning Materialization

Once constructed, the BIM is materialized as an executable logical representation. In this work, Prolog was selected as the primary reasoning substrate due to its declarative semantics, explicit logical inference model, and strong support for explainable symbolic reasoning (Wielemaker et al., 2012). The reconstructed semantic commitments are operationalized as logical predicates, taxonomic structures, evidential assertions, and inferential rules capable of supporting executable decision queries.

This transformation allows decision-support conditions originally expressed in informal business language to be evaluated through deterministic logical reasoning. Because the reasoning structures remain explicitly connected to their evidential origins through the BIM, the resulting decision process preserves traceability and explainability.

Importantly, the BIM is intentionally designed as a reusable semantic artifact. The ontology-guided unpacking process is executed once for a given business intent, while the resulting model can be reused across multiple downstream analytical reasoning cycles, enabling semantic consistency, operational efficiency, and future alignment with heterogeneous organizational data sources.

5 Architecture and Implementation

To operationalize the proposed methodology, we implemented a prototype decision-support pipeline in the context of cybersecurity risk assessment for organizational contracts. This scenario was selected as a representative case in which business-oriented textual artifacts, including policies, risk assessment guidelines, and domain questions, must be transformed into executable semantic structures capable of supporting explainable decision analysis. While the implementation is instantiated in this domain, the architecture was deliberately designed to preserve methodological generality.

Figure 1 reflects the operational realization of the proposed approach. The architecture follows four main stages: ingestion of heterogeneous business-oriented textual artifacts, ontology-guided AI-mediated semantic unpacking, construction of the Business Intent Model (BIM), and materialization of executable reasoning artifacts.

5.1 Canonical Semantic Compilation

A central implementation decision concerns the use of a canonical semantic schema for BIM construction. Rather than directly generating ad hoc domain predicates, the AI-mediated unpacking process first produces a normalized semantic representation that preserves structural consistency across inputs. This intermediate representation captures the main semantic categories aligned with the BIM structure, including concepts, entities, relations, constraints, taxonomies, decision rules, and question frames.

Listing 1: Excerpt of the canonical schema used for BIM construction

```
{
  "type": "object",
  "required": ["doc_meta", "concepts", "
      entities", "relations",
    "constraints", "taxonomies", "
        decision_rules", "question_frames"
  ],
  "properties": {
    "doc_meta": {"type": "object"},
    "concepts": {"type": "array"},
    "entities": {"type": "array"},
    "relations": {"type": "array"},
    "constraints": {"type": "array"},
    "taxonomies": {"type": "array"},
    "decision_rules": {"type": "array"},
    "question_frames": {"type": "array"}
  }
}
```

This canonical structure provides a stable semantic interface between heterogeneous textual inputs and downstream executable reasoning. It also enables structural validation of AI-generated outputs and improves reproducibility across repeated unpacking cycles.

5.2 AI-Mediated Ontological Unpacking

The semantic unpacking process is mediated by a Large Language Model acting under explicit extraction constraints. Rather than functioning as a direct answer generator, the model operates as a semantic compiler responsible for reconstructing ontological commitments embedded in business-oriented textual artifacts. To improve consistency and reduce speculative generation, the extraction process is governed by explicit semantic instructions aligned with the canonical BIM schema.

Listing 2: Excerpt of extraction instructions used in AI-mediated ontological unpacking

```
You are a semantic extractor specialized in
    ontological unpacking.

You will receive FREE TEXT written in natural
    language.
Your task is to return ONLY a strictly valid
    JSON object conforming
to the canonical schema provided.

Rules:
1. Do not invent information. Each extracted
    element must
   include evidence grounded in the source
       text.

2. If the text contains definitions, extract
    them as concepts.

3. If the text contains lists, levels, or
    classifications,
   extract them as taxonomies.

4. If the text contains conditional decision
    logic,
   extract it as decision_rules.

5. If the text contains explicit or implicit
    questions,
   extract them as question_frames.
```

This design enforces three important properties: semantic explicitness, structural consistency, and evidential traceability. As a result, the AI-mediated unpacking process systematically reconstructs the latent semantic commitments underlying the original busi-

Table 1: Representative semantic elements extracted during BIM construction.

Component	Example label/predicate	Description
Concept	cybersecurity risk	Risk associated with access to networks, critical infrastructure, or sensitive data
Evidence: "Cybersecurity risk is related to equipment connected to external networks."		
Entity	contract	Organizational artifact to be evaluated
Evidence: "Which contracts present cybersecurity risks?"		
Relation	access_to_internal_network	Possible relation between an external actor and the internal network
Evidence: "external companies may access the internal network."		
Taxonomy	service provider criticality	Ordered classification used to represent criticality levels
Evidence: "Service-provider criticality may be classified according to the following levels."		

ness intent rather than merely extracting isolated textual fragments.

5.3 Empirical BIM Construction

To illustrate the operational behavior of the architecture, we consider the cybersecurity decision-support scenario used throughout this paper. The input textual artifacts describe situations involving external access to organizational networks, handling of sensitive information, and service-provider criticality conditions. Although originally expressed in informal business language, these descriptions contain latent ontological commitments concerning entities, relations, evaluation criteria, and operational questions.

Table 1 presents representative semantic elements extracted during AI-mediated unpacking and normalized into the BIM structure.

Operational question structures are also explicitly preserved as part of the BIM, functioning similarly to competency questions in ontology engineering. These question frames define the decision-support queries that the resulting reasoning model should be capable of answering.

5.4 Executable Reasoning Materialization

Once the BIM is constructed, its reasoning layer is materialized as executable logical structures. In our implementation, Prolog was adopted as the reasoning substrate due to its declarative semantics, explicit inferential transparency, and suitability for explainable symbolic reasoning.

Listing 3: Excerpt of executable BIM materialization in Prolog

```
concept(cybersecurity_risk).

entity(contract).
entity(external_company).
entity(internal_network).

relation(access_to_internal_network,
        external_company,
        internal_network).

question_frame(criticality_assessment).

decision_rule(high_criticality) :-
    relation(access_to_internal_network,
            external_company,
            internal_network),
    access_to_sensitive_data(contract).
```

This materialization demonstrates that the proposed methodology produces executable semantic artifacts rather than merely descriptive conceptual structures. Because the generated reasoning rules remain explicitly grounded in the extracted BIM and linked to textual evidence, the resulting decision-support process preserves semantic traceability, interpretability, and reproducibility.

Although auxiliary representations such as JSON may support validation and interoperability, the central contribution of the implementation lies in demonstrating that AI-mediated ontological unpacking can systematically produce reusable executable reasoning artifacts suitable for grounded decision support.

Table 2: Representative question frames extracted from business intent.

Question	Intent
Will the service provider have access to sensitive information?	information_security
Does the service involve access to core IT systems?	information_security
Will contractors access internal IT assets?	information_security
Will personal data be shared with the provider?	lgpd

6 Evaluation

This evaluation assesses whether the proposed ontology-guided methodology effectively transforms semantically underspecified business intent into executable semantic structures suitable for grounded decision support. In alignment with the research questions introduced in the Introduction, the evaluation addresses three complementary dimensions: (RQ1) the semantic adequacy of the generated Business Intent Models (BIMs), (RQ2) the comparative characteristics of the proposed approach relative to a retrieval-based question-answering baseline, and (RQ3) the operational viability of executable reasoning over the resulting semantic artifacts.

The evaluation was conducted using the cybersecurity decision-support scenario introduced throughout this paper, involving the assessment of cybersecurity criticality in organizational contracts. The input corpus consisted of business-oriented textual artifacts describing cybersecurity assessment criteria, risk conditions, and operational decision requirements. To establish an expert reference model, three domain specialists independently analyzed the same source material and manually constructed reference BIMs according to the canonical semantic categories adopted by the methodology: concepts, entities, relations, constraints, taxonomies, decision rules, and question frames. Divergences among expert annotations were subsequently resolved through consensus reconciliation, producing the final expert reference used in the qualitative assessment.

As comparative baseline for RQ2, we adopt the retrieval-based enterprise question-answering architecture presented in REF1, which addresses the same organizational contract domain through document retrieval, semantic similarity, and LLM-mediated answer generation rather than ontology-guided semantic compilation. This baseline provides an appropriate contrast because it represents a contemporary retrieval-centric paradigm for enterprise AI decision support.

6.1 RQ1: Semantic Adequacy of BIM Construction

The first evaluation dimension examines whether the AI-mediated unpacking process produces semantically adequate BIMs relative to the expert reference model. Rather than relying on purely lexical matching, the assessment considered semantic equivalence under the canonical BIM categories, focusing on whether the generated structures preserved the decision-relevant ontological commitments identified by the specialists.

Table 3 summarizes the qualitative assessment.

The results indicate that the proposed methodology performs particularly well for semantically explicit categories such as concepts, entities, and question frames. These structures are more directly verbalized in business-oriented textual artifacts, making their ontological commitments easier to reconstruct under constrained AI mediation. Relations and constraints also demonstrated strong performance, suggesting that the methodology effectively captures semantically relevant dependencies even when only partially explicit.

More structurally complex elements, particularly taxonomies and decision rules, exhibited lower completeness. These structures often depend on reconstructing distributed evaluative logic rather than extracting isolated semantic statements. Even in such cases, however, the generated BIMs consistently preserved the minimum semantic distinctions necessary for executable reasoning. From a decision-support perspective, this is particularly relevant because the objective is not merely semantic extraction, but the reconstruction of decision-relevant formal structures.

6.2 RQ2: Comparison with Retrieval-Based Enterprise QA

The second evaluation dimension compares the proposed BIM-based methodology with the retrieval-centric architecture adopted in REF1.

The comparison highlights a fundamental architectural distinction. REF1 excels in flexible information access across heterogeneous enterprise sources,

Table 3: Qualitative assessment of BIM construction against expert reference.

Semantic category	Assessment	Observations
Explicit concepts	Strong	Consistently extracted from directly stated domain definitions.
Entities	Strong	High semantic correspondence with the expert reference model.
Relations	Moderate–Strong	Relevant semantic dependencies were generally reconstructed when partially explicit.
Constraints	Moderate–Strong	Evaluative restrictions and assessment conditions were adequately preserved.
Taxonomies	Moderate	Hierarchical semantic distinctions were identified, though some deeper structures were omitted.
Decision rules	Moderate	Conditional assessment logic was partially reconstructed, especially when distributed across textual fragments.
Question frames	Strong	Operational decision-support questions were consistently preserved and formalized.

but its reasoning remains fundamentally retrieval-mediated and probabilistic. In contrast, the methodology proposed in this paper introduces an explicit semantic compilation stage in which business intent is transformed into executable formal structures before downstream reasoning occurs. This difference is particularly significant for decision-support scenarios requiring reproducibility, formal traceability, and inspectable reasoning behavior.

6.3 RQ3: Executable Reasoning Viability

The third evaluation dimension examines whether the generated BIMs support executable decision reasoning in practice. To assess this capability, the cybersecurity scenario was operationalized as an executable decision query over the generated Prolog reasoning layer.

A representative example is the following executable decision-support query:

Listing 4: Representative executable decision-support query

```
?- assess_cybersecurity_criticality(
    contract_050_2021, CriticalityLevel).
```

This query evaluates cybersecurity criticality by applying the decision structures reconstructed during ontology-guided unpacking, including conditions related to access to sensitive information, access to internal infrastructure, and operational risk dependencies.

Unlike retrieval-based architectures, which generate textual responses probabilistically based on retrieved context, the proposed approach performs deterministic symbolic inference over explicitly modeled semantic structures. Because the reasoning process operates over inspectable predicates and rules derived from the BIM, every decision outcome remains traceable to explicit semantic commitments and supporting evidence.

This capability demonstrates that the proposed methodology does not merely generate descriptive semantic artifacts, but produces operational reasoning structures suitable for grounded decision support.

6.4 Operational Feasibility and Scalability

From an operational perspective, the ontology-guided unpacking process is executed only once for a given business intent, while the resulting BIM remains reusable across multiple downstream reasoning cycles. This architectural separation substantially reduces recurring computational costs.

In the evaluated cybersecurity case, a source corpus of approximately 1,300 words produced a BIM containing 169 executable Prolog lines and 144 structured predicates spanning concepts, taxonomies, decision rules, evidential links, and operational question frames. Using Google Gemini 2.5 Pro, the complete unpacking cycle consumed approximately 2,200 input tokens and 3,800 output tokens, corresponding to an estimated generation cost of approximately US$0.05 per BIM.

To assess scalability, reasoning execution was tested under synthetic replication of the generated BIM structures. The symbolic reasoning layer exhibited predictable performance growth as model size increased, reflecting the deterministic behavior expected from declarative logical execution. While broader scalability evaluation across heterogeneous enterprise domains remains future work, these results indicate that the proposed methodology is operationally viable for realistic decision-support scenarios.

Table 4: Conceptual comparison between the proposed BIM-based methodology and the retrieval-based baseline (REF1).

Analytical aspect	REF1 Retrieval-Based QA	Proposed BIM-Based Methodology
Knowledge representation	Implicit document chunks and semantic embeddings	Explicit executable semantic model
Semantic explicitness	Latent similarity-based representations	Explicit ontological commitments
Decision criteria formalization	Embedded in prompts or retrieved context	Explicitly modeled as formal decision structures
Traceability	Textual justification from retrieved passages	Evidence-linked semantic commitments with explicit grounding
Reproducibility	Dependent on retrieval context and generative variability	Deterministic symbolic reasoning over executable artifacts
Explainability	Post-hoc natural-language explanation	Inspectable predicates, rules, and semantic dependencies
Executable reasoning	Not natively supported	Native logical execution over BIM reasoning structures
Suitability for grounded decision support	Contextually informative but probabilistic	Structurally grounded and decision-oriented

7 Conclusions and Future Work

This paper introduced an ontology-guided methodology for *Business Intent Unpacking*, aimed at transforming semantically underspecified business-oriented textual artifacts into explicit executable semantic structures for grounded decision support. Building on ontology-driven conceptual modeling and ontological unpacking, the proposed approach formalizes business intent through the *Business Intent Model (BIM)*, a multilayer semantic artifact that captures intent semantics, structural ontological commitments, evidential grounding, and executable reasoning structures. By treating AI as a semantic compilation mechanism rather than a direct answer generator, the methodology enables the systematic transformation of informal business intent into inspectable reasoning artifacts.

The evaluation provided evidence along three complementary dimensions. Qualitative expert assessment indicated that the methodology consistently preserves the semantic distinctions most relevant to decision-support scenarios, particularly for concepts, entities, constraints, and operational question structures. Comparison with the retrieval-based baseline highlighted a fundamental architectural distinction: while retrieval-centric approaches provide flexible contextual access to enterprise information, the proposed methodology produces explicit executable semantic structures that support deterministic, traceable, and explainable reasoning. Operational evaluation further showed that BIM construction and execution are computationally feasible in realistic cybersecurity criticality assessment scenarios.

More broadly, this work contributes to connecting research streams that have often evolved independently, including ontology-driven conceptual modeling, requirements engineering, ontology learning, neuro-symbolic reasoning, and explainable enterprise AI. The central contribution lies in introducing an explicit semantic compilation stage between informal business-oriented artifacts and downstream analytical reasoning, enabling decision-support systems to operate over formalized semantic commitments rather than relying exclusively on retrieval-mediated generative inference.

A primary direction for future work concerns the establishment of a *Semantic Bridging* layer capable of aligning generated BIMs with heterogeneous organizational knowledge substrates, including relational databases, knowledge graphs, operational systems, and regulatory repositories. Such alignment is essential for enabling grounded reasoning over factual organizational evidence in real enterprise settings. Additional future work includes improving the robustness of AI-mediated unpacking over larger heterogeneous corpora, extending the methodology to more complex multi-criteria decision scenarios, and incorporating interactive clarification mechanisms for progressively refining incomplete business intent.

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